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A systematic literature review of methods to describe the stereo image of recorded music

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Abstract

Analysing spatial information within recorded music is increasingly important for audio research given the explosion of available datasets, but relevant techniques are spread over diverse fields. To address this, a systematic literature review (SLR) was conducted to ascertain all existing methods for describing the spatial features of stereophonic music. The Scopus, Web Of Science (WOS), and Semantic Scholar (SS) databases were searched on 26th February 2026, and articles were deemed eligible if the method they described included any extraction of spatial information from a music audio signal. Links and similarities were drawn between studies to identify the core methods of describing stereo image. One hundred seventy-eight articles were included, with 29 of those deemed "very relevant". The fundamental methods of extracting spatial information are detailed and a framework for creating a custom stereo image profile (SIP) is drawn up, followed by suggestions for future research. The review was limited by the substantial volume of literature to be assessed by few researchers, although this wide breadth served to mitigate bias by gathering at least one example of all existing SIP methods. Inconsistent application of terminology within the existing literature presented a significant challenge. The work was funded in part by The Knowledge Foundation (KKS).

Keywords Music information retrieval, Stereo image, Systematic literature review

1 Introduction

The extraction of spatial information from musical audio signals has long been a goal for digital music practitioners, with downstream uses such as classification, recommendation, format conversion, and even musicology. However, until the recent development of object-based audio, it was not common practice to include any such information in music metadata, and it must therefore be derived after-the-fact from the final musical mix. With the new availability of very large musical datasets there exists an opportunity to gain insights into musicians' use of spaciousness in their work.

It is imperative to recognize that the gamut of spatial processing in audio research is enormous, and the terminology, keywords, and article structure somewhat non-standardized. Many studies include the extraction of spatial information in their methods, and while said information may be enough to describe stereo field, the articles are written from the perspective of the downstream task rather than the spatial features themselves. Consequently, *very relevant* literature is difficult to find using simple searches, yet increasing the search radius results in excessive amounts of *semi-relevant* literature. This represents a missed opportunity for "cross-pollination" between adjacent (but usually independent) audio research fields, and to that end, this SLR aims to determine all currently available methods to describe the stereo field of recorded music. It follows the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines [1].

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1.1 Previous reviews

Reviews of techniques to extract spatial information from audio are generally rare, with the few examples being tailored to specific use cases. For example, [2] gives an overview of machine learning (ML)-based techniques for the capture, processing, and rendering of spatial audio. The targeted use cases are gaming, augmented reality (AR)/virtual reality (VR), teleconferencing, and entertainment applications. While binaural audio is mentioned as a spatial format, other stereophonic recordings are not, nor is the focus on musical audio. The authors also state that their review is "representative... [but not] exhaustive". Most interestingly, they conclude that tasks such as "analysing the spatial properties of sound [had] not been... disrupted by ML methods". In a similar vein, [3] is an SLR of the technologies and applications of spatial audio in AR. The authors state that it is the first systematic review in the field, but the audio sections are focused on synthesis rather than analysis, and (like the former) musical audio is not targeted. A more recent review [4] focused specifically on spatial audio, mentioning music as a use case. It too only references binaural audio (and not stereophonic more generally) as a spatial format, and is focused on research performed from 2022 onwards. It does, however, give an excellent overview of input and output audio representations, spatial audio tasks, available datasets, and evaluation metrics.

1.2 New aims and scope of this review

To fill gaps in previous reviews, this SLR focuses specifically on techniques to describe the spatial features of stereophonic musical audio, extracting them from previous literature and gathering them in a single place for reference by the audio research community. The aforementioned lack of standardization will be normalized such that future researchers can share a common footing and language. See Sect. 2.5 for a detailed explanation of specific outcomes.

A list of sonic qualities that represent audio "spaciousness" is defined in [5], from which it is possible to infer that any audio signal with more than one channel is eligible to contain spatial information. Rumsey [6] states that even though it is possible to perceive distance (which can be considered a spatial attribute) in a monophonic recording by assessing the relative loudness of sources or extent of reverberation, *width* is a far more prominent spatial attribute that is impossible to reproduce monophonically. To keep the scope of this SLR reasonable, monophonic audio is therefore deemed out of scope despite the fact the assessed techniques would be inapplicable to older and historical recording collections. Conversely, while many studies act on audio with more than two channels, they are still deemed relevant if the underlying techniques could be applied to stereophonic audio.

2 Methods

This section defines all processes for performing the review, and discusses some known limitations.

2.1 Search process

The review follows the Population, Context, Outcome (PCO) protocol [7]. The population is defined as "music", the context as "stereo", and the outcome as "spatial information". Since these terms and their synonyms are often not standardized, a list of negative search terms were also included to exclude some obvious false positives. Several very specific terms that were already known to the author were also included, and any article using those terms was included in the initial search result. The full lists of synonyms, specific terms, and negative terms used to form search queries, plus the search queries themselves can be found in Appendix 1. The search was performed on 26th February 2026.

Table 1 shows the databases that were searched, and their precedence. Both Scopus and WOS have a browser-based search functionality accepting advanced queries, from which the results can be exported as Comma separated variable (CSV)/Tab separated variable (TSV) files. SS provides an application programming interface (API) endpoint that can be used to perform similar advanced searches [11]. It was found during initial searches that although the SS API accepts negative terms, it often wrongly excluded relevant articles. Negative terms were therefore omitted from the SS query. The data exported from the searches was universal resource locator (URL), title, abstract, authors, publication year, publication source, Digital Object Identifier (DOI), and citation count of each study. A full dataset of the collected search

Table 1 Databases used for article search

	Database	Description
1	Scopus [8]	The largest database of scientific articles, indexing a majority of computer-science publishers.
2	Web Of Science [9]	Another well-respected general scientific database that acts to fill gaps in Scopus.
3	Semantic Scholar [10]	A general-purpose scientific database created via web-scraping that tends to surface more "grey" publications such as theses, commercial white papers, and pre-prints, but can be prone to errors and false records.

results for all databases is included with this manuscript as an additional file.

2.2 Initial screening

The initial list of articles was filtered to remove duplicates, firstly identifying them by DOI, and then by title and publication year when DOI was not available. Since there are can be differences in formatting between the databases, the titles were first "normalized" by converting to lower case and removing all non-alphanumeric characters. Articles were marked as duplicates if the normalized title and publication year exactly matched one previously seen in the list. See Appendix 4 for pseudocode that describes the automatic filtering process.

A first pass was then performed to make obvious exclusions. The title and abstract of each paper was read, and the article was rejected if any of the following were not true:

- Database record was valid.
- Title and abstract were written in English.
- Document was available for reading.
- Document described some research[1].
- Subject of the study was audio.
- Described method included the extraction of spatial features from an audio signal at any point.

A record was considered valid if it linked to a published document. In practice, invalid records were only found in the SS database, which (as previously mentioned) is created via web scraping and is hence prone to errors. For example, SS record 196093012 was included in the search results, but is considered invalid. Some duplicates were also identified manually and rejected at this stage. Note that the conditions of the audio comprising at least two channels and being musical were not assessed during the screening as it was very often not obvious from the title and abstract. If the researcher was unsure about whether an article should be rejected or not, it was marked as "unclear" and revisited at a later stage.

2.3 Review process

2.3.1 Primary review

After the initial pass, all non-rejected articles and all articles marked as "unclear" with a publication year of 2020 or later were obtained and read in full. They were assessed (and possibly rejected) based on the same criteria mentioned in Sect. 2.2, plus two additional conditions:

- The audio signal comprised at least two channels
- The audio signal was musical

To assess whether an audio signal is musical or not, it must meet the condition that it contains a recording of at least one musical instrument (including singing voice), or that the study specifically stated that it was applicable to music. Studies using both musical and non-musical signals were also included.

While it is common to exclude reports based on the impact factor or citation count [12], an initial scan of the search results revealed that some very obviously relevant studies had very few, or even zero citations (e.g., [13]) or were published as pre-prints only (e.g., [14]). Therefore, citation count and publication venue were not taken into account.

2.3.2 Post-hoc screening test

As a post-hoc test, a second researcher was given a random 5% sample of the search results (with automated rejections already removed) and independently screened them based on the conditions specified in Sects. 2.2 and 2.3.1. Discrepancies between the researchers' opinions were flagged for discussion only if the second researcher included a study that the first had previously rejected. Table 2 shows the results, and ultimately, only one such study [15] was found. The first researcher marked it as "unclear" and later rejected it because the publishing year was missing from the database search result. After reading the full paper both researchers agreed to reject it because it did not describe the extraction of any spatial features. Overall, the post-hoc test did not raise any issues with the primary review.

2.3.3 "Very relevant" studies

Since the number of relevant studies from the initial assessment was high, a further selection process labelled those deemed "very relevant" to the aims described in Sect. 1.2. This process filtered studies based on their "downstream task" (see Table 3), selecting tasks that are relevant to the extraction or interpretation of spatial information in large audio corpora. As per the PRISMA

Table 2 Results of post-hoc screening test

Outcome	Count	Treatment
Initially rejected, post-hoc included	1	Researchers discuss
Both included	3	OK
Initially included, post-hoc rejected	13	OK
Rejected for same reason	68	OK
Rejected for different reasons	68	OK
Total	153	–

[1] Other document types were e.g. book review, patents, contents of conference proceedings, etc.

Table 3 Data extracted from reviewed articles

Data	Description	Examples
Audio model	The channel count or standard paradigm that the audio signal follows	stereo, binaural, ambisonic, mic array
Contents of signal	A high-level description of what the audio contained	music, speech, noise
Specific audio configurations	More detail about the signal contents or how it was recorded/created	reverberant, synthetic binaural, orchestral
Datasets	Any audio datasets that were used in the creation of the audio signal	FMA, MTGJ, MedleyDB
Downstream tasks	The specific tasks that the spatial information was used (or was intended to be used) for	Source separation, MIR, classification, compression
Novel mechanism category	A high-level category of the articles main contribution	DSP, ML, psychoacoustic test
Spatial attributes	The specific spatial information that was extracted from the audio signal	DOA, panning histogram, cochleagram
Novelty	A record of whether the article describes a new spatial feature or referenced one from the literature	novel, referenced, modified
Evaluation technique	The technique by which the scientific claims were verified	Objective metrics, subjective test, simulations

guidelines, those studies are listed and summarized in Sect. 3.2 — Readers may use them as a main reference point for existing techniques that directly relate to the aims stated in Sect. 1.2. All studies are still deemed "relevant", forming the basis for the syntheses described in Sect. 2.5, and providing the building blocks for readers to construct spatial descriptors specific to their task.

2.4 Data extraction

For each study included in the review, data was collected according to Table 3. Note that the options in each category are not mutually exclusive, e.g., [16] describes Binaural Cue Coding (BCC) which, while originally intended for the compression of multichannel signals, is also used for stereo. An additional example is [17] which states that its downstream task is music information retrieval (MIR), and then provides proof of the method's effectiveness on a classification task.

2.5 Syntheses

The primary synthesis is an exhaustive list of methods to generate a SIP for musical audio, plus a thorough analysis of their underlying extraction techniques, and what musical features they are able to detect. Complementary to that is an explanation and normalization of their terminology, drawing links where studies are using similar techniques and differentiating between homonymous terms.

Following that, a framework of how to choose or form a SIP that is suitable for any particular task. The data extraction process differentiates between traditional digital signal processing (DSP) and newer ML/deep learning (DL) spatial features, and it will be noted which tasks are currently benefiting from modern techniques, thereby giving inspiration for how to incorporate them in other tasks.

In addition, a list of useful audio datasets and experiment validation techniques will be formed so that any future researchers can find tools suitable for their tasks. Propositions for future research will also be given based on what is missing from the search results.

2.6 Limitations

Several limitations were uncovered during the review process. Firstly, due to the wide research questions and non-standardized formats of audio-related article abstracts, the number of articles requiring review was very high. Even though a second researcher performed a post-hoc test to judge article screening, only a single researcher was tasked with extracting data from all articles. This introduces a potential source of bias. However, the review's wide breadth was also a point of strength since the aim was to gather at least one example of all existing SIP algorithms. It is difficult to assess the impact of audio processing-related articles since they typically do not receive many citations and the publication venue and other related information is often missing from the databases. The solution was to assess each article equally, which created more manual work.

In addition, the non-standardized terminology meant that the database search queries (see Appendix 2) contain many synonyms for "spatial information". The list was built iteratively over several initial searches, but there is still a chance that some missing keywords would have been beneficial to include. However, since this review is not concerned with the absolute number of search results for each term, this limitation is largely mitigated as long as at least one example of any given spatial descriptor was

present in the results. Similarly, while the search process included specific terms for commercial spatial audio formats, information about them (and their underlying algorithms) is typically not public and hence was absent from the search results.

There is likely a bias regarding the publication venue of the articles sought for retrieval since some specific journals were not available through library subscriptions[2]. Articles appearing potentially relevant were sought out via other methods (internet searches or direct contact with the authors) as a mitigation. Several doctoral theses were surfaced in the search results, and while they were relevant they were difficult to categorize in the review's framework since they typically contain multiple studies and hence fall into multiple categories. The accuracy of the data extracted from them could therefore be unreliable.

The assessment of DL-related papers presented a few challenges. Firstly it is quite common to simply cite the audio dataset used, rather than describing the format and contents of the audio. Additionally, due to the black-box nature of DL, the trained network often *does* learn spatial features, but it is debatable whether that constitutes a spatial description (see Sect. 3.1 for examples).

3 Results

The search initially identified 4678 studies, of which 1548 were removed before screening. Two thousand three hundred sixty-seven were excluded after reading the title and abstract, leaving 763 that were sought for retrieval. Out of those, 48 could not be retrieved and 537 were rejected, leaving 178 that were included in the review. Twenty-nine of those were deemed "very relevant" due to their downstream task. Using those 178 studies as a seed, 15 studies were added by following references, 4 studies were found using the SS recommendation service, and 5 studies were recommended by peers. A full flow diagram of study inclusion and rejection is shown in Fig. 1. In total, 14 studies were pre-prints and had not seen peer review at the time of the search or recommendation.

As mentioned in Sect. 2.3, some studies were marked as "unclear" during the initial screening. As a metric for the judgement of the researcher, the number of studies assessed and eventually included are shown in Table 4. It shows that a far lower proportion of "unclear" studies were eventually included, implying that the researcher's judgement was largely correct.

[2] The most prominent unavailable journals were "Proceedings of Forum Acusticum", "Proceedings of the International Congress on Acoustics", and "Journal of the Acoustical Society of America".

3.1 Borderline studies

Some studies presented edge cases during assessment. The following non-exhaustive list gives some examples of such studies, explaining why they were either included or excluded.

3.1.1 Inclusions

- Pollow et al. [18] describes a method that acts on audio from a large microphone array rather than recorded music, but it is included because it mentions the nuances of using phase and intensity differences to estimate sound directivity.
- Bourbon et al. [19] uses multichannel renderings of object-based audio as the source, but was included because the method acts on pairs of channels as opposed to all channels simultaneously.
- Fernandez et al. [20] is primarily focused on the rendering of binaural microphone array signals, but was included because it references several useful algorithms for calculating spatial attributes of audio.

3.1.2 Exclusions

- Lindau et al. [21] contains a useful list of spatial audio terms, but does not describe any audio processing.
- Colone and Reiss [22] describes a method for predicting panning positions of stems in a stereo mix, but required access to those stems in order to function.
- Felipe et al. [23] uses stereo features to aid in the classification of binaural scenes, but does so by extracting monophonic features from each channel.
- Sierra et al. [24] is focused on capturing spatial audio for MIR purposes, but simply outlines the capture process and choice of spatial format rather than deriving any spatial information.
- Pulkki et al. [25] is a software implementation of the underlying spatial processing technique, which is described academically elsewhere in the literature.

3.2 Very relevant studies

This section will highlight the studies that are very relevant to the review's aims given in Sect. 1.2, summarize each study, and draw links between them. Generally speaking, very relevant tasks are those that relate to creating a high-level spatial description of a recording, or describing the spatial features of large musical corpora. The tasks deemed not very relevant are more concerned

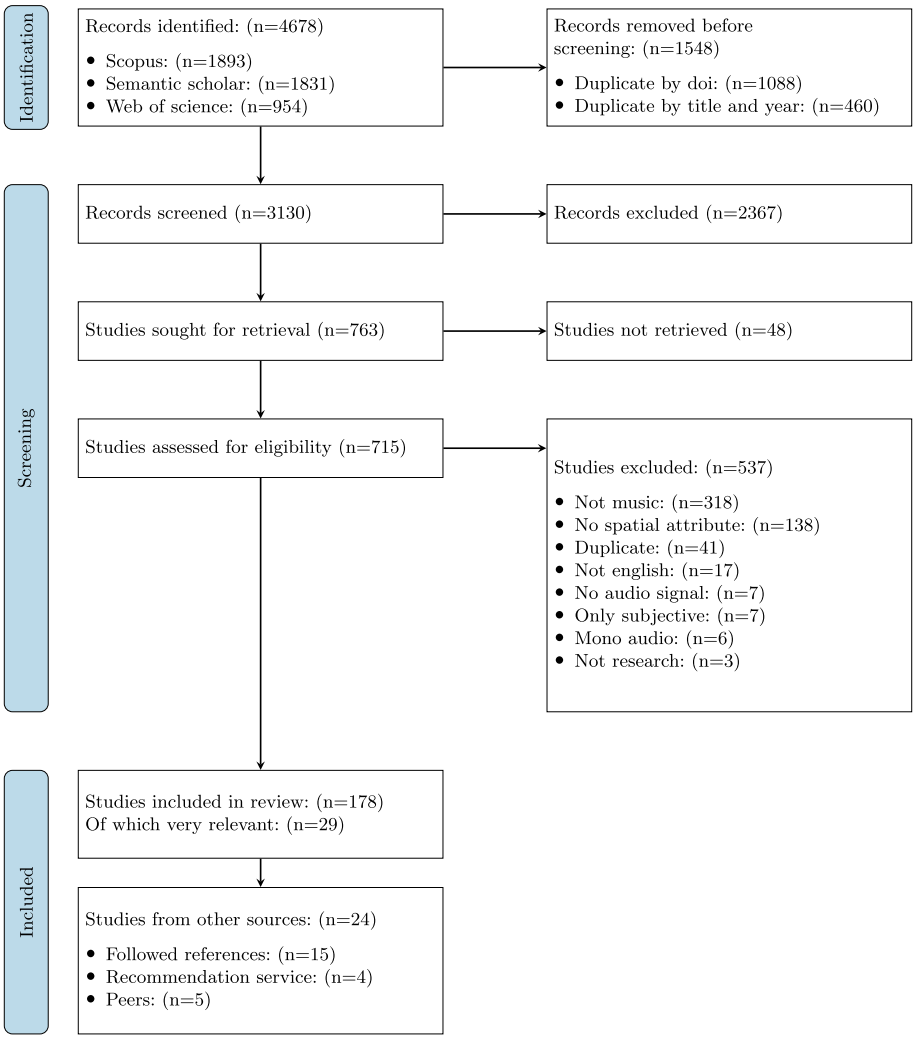


Fig. 1 PRISMA flow diagram

Table 4 Proportions of studies that were included in the review, split by clarity

	Assessed	Included	Proportion
Obvious	631	175	28%
Unclear	84	3	4%
Total	715	178	25%

with low-level spatial features and often require the audio to be reconstructed or altered after analysis. A full list of downstream tasks can be found in Appendix 5. In this SLR the 29 studies deemed to be very relevant are those whose downstream task was at least one of "MIR", "spatial analysis", "musicology", "visualization", or "classification". These studies tend to fall into tightly-related cohorts, often referencing each other.

The earliest is [26]—a doctoral thesis defining methods for spatial analysis of binaural signals for use in the broadcasting industry. Dewhirst [27] (another doctoral thesis) expands on [26], combining it with a binaural hearing model and an interaural cross correlation (IACC)-based stereo width estimator [28] to create an artificial listener with the intention of assessing perceived sound at arbitrary (i.e., non-sweet-spot) listening positions.

A method to estimate amplitude panning position in the time-frequency domain (named panning index (PI)) was defined in [29, 30] and demonstrated for upmix and source separation tasks. Tzanetakis et al. [31] and its extension [17] define the stereo panning features (SPF)—a descriptor of amplitude panning based on the PI (although renamed to stereo panning spectrum (SPS)). In turn, [32] uses SPF as a measure of similarity between two mixes of the same musical stems, and [33] used the

SPS to visualize artistic audio recordings and SPF as a measure of similarity between those recordings.

Sarroff and Bello [34] defines a spatial descriptor named panning histogram (PH), which is based on the Frequency-Azimuth (FA) plane defined in [35]. The PH is also collapsed into a single measure of "stereo width", which the same author uses in [36, 37] to create a computational model of perceived spaciousness. Notably, these studies also use monophonic descriptors, including a prediction of the extent of reverberation in the recording.

The concept of a PH is used by several other studies, although under different names and with different mathematical definitions. Takanen and Lorho [38] generates likelihood histograms of the direction of arrival (DOA) of sound sources from measurements of the interaural level difference (ILD) and interaural time difference (ITD) in binaural correlograms, from which a measurement of stereo width is also derived. A major difference from the previously mentioned works is that the method requires calibration using a recording of background noise, and reference ILD/ITD values from which the DOA is calculated. Gomez et al. [39] defines another PH which is based on the PI from [29]. Furthermore, it defines a method of PH quantization with the intention of allowing simple similarity measurement of music recordings in a lower-dimensional, non-perceptual domain. Differing slightly from other studies, [40] focuses mostly on the visualization of stereo audio. It transforms time-frequency (TF) domain interaural intensity difference (IID) and interaural phase difference (IPD) into two cues — "position" and "clarity" — which are then used to create several visualizations including a PH. The same author expanded on this idea in [41], defining a transformation from time-domain stereophonic signal into a generic bivariate mixture space (BMS), which in turn is used for visualization and panning estimation (among others).

Tardieu et al. [42] provides two alternative descriptors of panning features—amplitude stereo features (ASF) for amplitude panning and phase stereo features (PSF) for phase panning. They are created from time-domain cochleagrams and frequency-domain spectrograms respectively, and are demonstrated on a decade classification task.

Härmä [43, 44] collectively demonstrate a new paradigm of SIP classification. As a precursor, a novel signal model is introduced that defines seven classes of stereo field. Subsequently, it is shown audio clips falling into six of the seven classes can be identified, first using a feed-forward neural network and then purely analytically. The inputs to the classifiers are cross correlations between the two channels and their variances (in TF domain). The downstream tasks are the aforementioned classification itself, as well as audio enhancement, audio manipulation, and upmixing.

You and Cheng [45] presents a method of comparing a degraded signal with its original in the spatial domain. It uses differences in IID and ITD within audio sub-bands, the primary use case being objective assessment of spatial compression algorithms. With a similar use case of predicting the perceptual quality of binaural renderings of microphone arrays, [46] uses both textural features defined by the Objective Measurements of Perceived Audio Quality (PEAQ) standard [47] and spatial features defined by its multichannel extension PEAQ-MC [48]. The spatial measures of PEAQ-MC are differences in ILD, ITD, and IACC between an original and degraded signal.

Chen and Hasegawa-Johnson [49] exploits the common mixing practice of contrasting highly centred sources with panned sources to tackle the task of speech-to-music-ratio prediction. After converting a stereo signal to mid/side configuration, a metric representing the ratio of centred to panned audio energy is defined. Somewhat similarly, [50] uses mid/side encoding of Mel-spectrograms to aid a DL classification of specific classes of binaural music recordings. Importantly, both studies perform the conversion to mid/side format in the time domain and use the resulting signal to construct features for input into an ML model.

Focusing on the enabling of new musical production techniques, [51] used analysis of goniometers and a derived measurement of stereo width to demonstrate the effectiveness of "dynamic polar patterns" in stereo microphone recordings. In a similar vein, [14] tackles a genre classification task based on measurements derived from virtual goniometers and channel correlation. Both studies use the goniometer output as an analogue to stereo width, noting that since mixing engineers refer to such tools during their work, there must be value in their readings. Regrettably, neither study considers that a goniometer's output inherently entangles phase and amplitude such that a gain alteration might induce a stereo width change, and vice versa.

Another estimator of stereo ensemble width is given in [52], based on "spatiograms" (originally defined in [53]) derived from a binaural signal, and the same author provides a comparison of different ensemble width estimators in [54]. Interestingly, spatiograms were originally intended to estimate DOA, are derived from the ITD, and are hence "phase-only".

Acting on stereo stems that have been separated from a mix using DL, [55] uses ILD to form panning maps of individual musical components, subsequently comparing songs from different decades and finding trends in music production. ILD is also used in [56] to determine the

amplitude panning azimuth of individual notes in recordings of polyphonic music, with the caveat that their fundamental frequencies have been determined beforehand.

In somewhat of a departure from the other studies surfaced in the review, [13] (while initially studying a DL upmixing task) demonstrated that the latent space learned by the neural network successfully disentangled spatial information from musical information, thereby enabling a vector within that space to represent the spatial qualities of the input audio. In a similarly unique task space, [57] uses the classification of stereo features to detect audio forgeries, although the features used are not learned, but rather are derived from classical DSP spatial cues such as ILD and IPD.

3.3 Fundamental concepts and terms

Many of the studies gathered in this review make use of similar fundamental audio concepts and signal processing techniques. The naming conventions are somewhat fluid, with a single term often used for variations of an underlying concept, or a single concept being referred to by multiple terms. This section aims to clarify these ambiguities.

3.3.1 Binaural and spatial cues

The terms "binaural cues" and "spatial cues" are umbrellas referring to any low-level spatial information extracted from two or more audio channels. They are used very broadly and do not have specific definitions, although some subcategories are common. Among the most used are interaural level difference (ILD) and interaural time difference (ITD), which are usually (although not always) extracted from frequency subbands of a stereo audio signal. They have previously been used in the context of psychoacoustic experiments as ways to explain and quantify the perceived effect of lateral reflections in acoustic spaces [58, 59]. Even though they contain the word "interaural" (Latin for "between the ears") they are freely used throughout the literature to mean amplitude or time difference between any two channels of an audio recording, e.g., binaural [60], stereo [45], and multichannel [61] audio. Barry and Lawlor [35] uses ILD as a ratio instead of its absolute form and [62] coins the term interaural level ratio (ILR) to formalize this, noting that since ratios are bounded in the range -1 to 1 they are usable as inputs for ML tasks. (In the classical form, ILD is usually given in decibels and is theoretically unbounded.)

The terms often appear with slight naming variations—synonyms for ILD include interaural intensity difference (IID) [26, 27, 63, 64], inter-channel level difference (ICLD) [65–67], and channel level difference (CLD) [68]. Likewise, a common synonym for ITD is inter-channel time difference (ICTD) [69–71]. Another formulation of ITD is IPD [20, 72, 73] (or inter-channel phase difference (ICPD) [74]) which is a frequency-dependent time difference. Notably, ITD can be calculated without knowledge of a center-frequency, as in [45] which uses transients instead.

Another common family of cues is IACC [46, 60, 75, 76], inter-channel correlation (ICC) [77, 78], inter-channel coherence (IC) [73], or interaural coherence (IAC) [79]. These cues indicate the correlation of 2 audio channels and are closely related to a perceptual audio property–listener envelopment (LEV). Faller and Merimaa [79] notes that IAC can be used to determine whether ITD and ILD represent a strong directional or more reflectional audio source. De Sena et al. [80] provides a similar perceptual measure of "location uncertainty". On a cautionary note, [76] mentions that IACC values measured from the same acoustical space are affected by the underlying musical material.

In the classical perceptual model of directional hearing it is accepted that IPD is only relevant for frequencies under 1500 Hz [81, 82]. However, there are several examples in the literature of IPD for all frequency bands being treated equally [40, 42, 83]. One reason for this is that while the directionality of a single narrowband source cannot be determined by a human using phase differences above 1500 Hz, reverberant or decorrelated sources often result in a broadband mixture of phase differences [84].

Importantly, there are various ways to calculate all these parameters, and [85] even notes that different definitions result in binaural/spatial cues with different underlying properties, hence subject to variation from different influences. For example [86, 87] both use the term ILD but calculate it in the time and time-frequency domains respectively. For an excellent overview of several classic methods for calculating ILD, ITD, and IACC the reader is directed to [26], and block diagrams showing some generic methods are shown in Fig. 2.

As a rule of thumb, it is advisable that when using any binaural or spatial cue for research, the author should preferably state an exact mathematical definition directly in the text, otherwise referring to a specific definition from another publication. In particular, it is advised that authors should use "interaural" terms for binaural audio only, opting for "inter-channel" in other cases.

3.3.2 Panning parameters

Another cohort of algorithms is dedicated to estimating the panning that has been applied to monophonic stems in a stereo mix. Those related to amplitude panning differ from the binaural ILD in that they account for a standard signal model in which the amplitude of the two channels follow the "energy-preserving panning law" whereby the

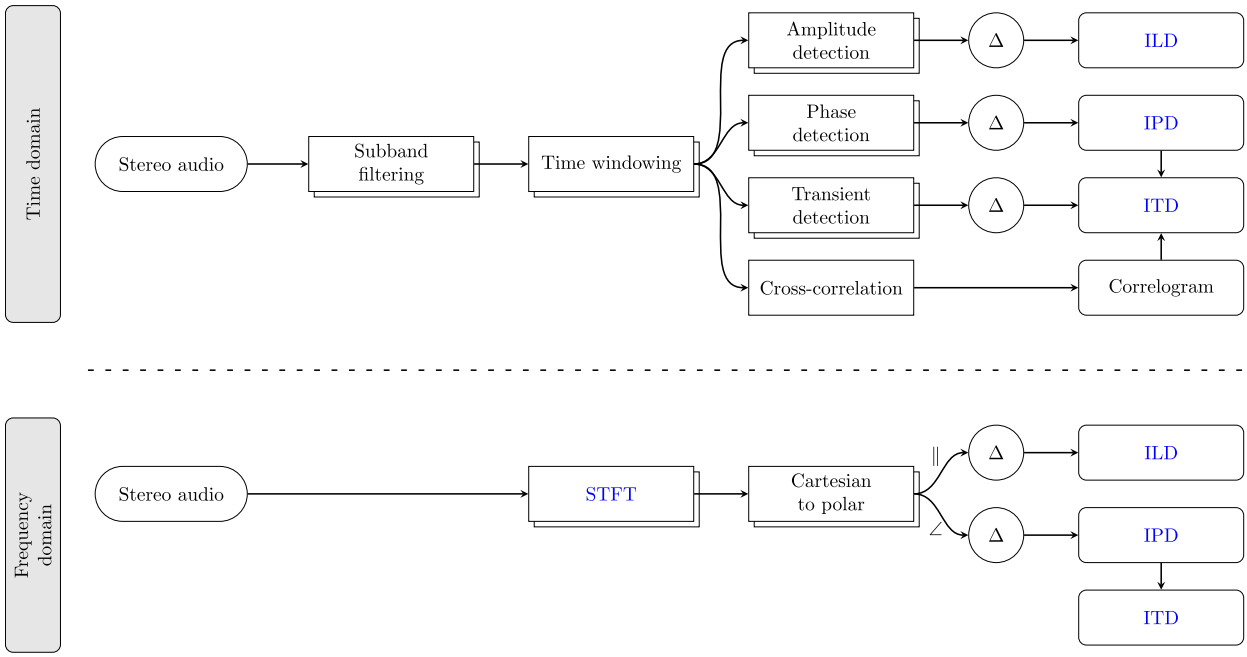


Fig. 2 Block diagrams of common spatial cues. Δ represents the inter-channel difference. ‖ and ∠ represent the magnitude and phase components of the short-time Fourier transform (STFT) respectively

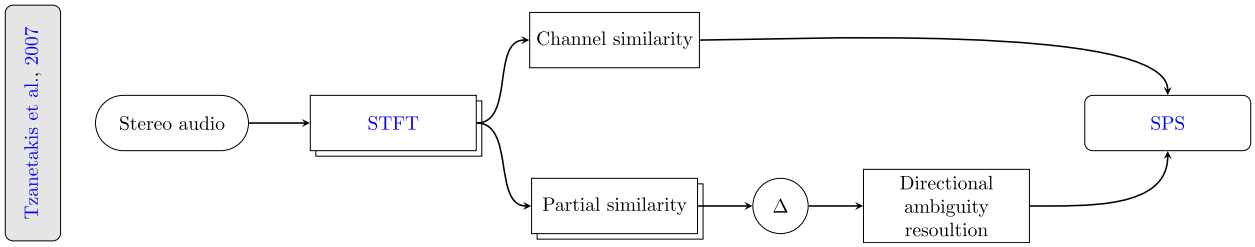


Fig. 3 Block diagram of SPS [31]. Δ represents the inter-channel difference

sum of the squared channel gains is 1. A formal mathematical formulation is given in [29, 30, 88] under the name panning index (PI), and [89–91] use the same technique unaltered. Similar formulations are found under multiple names: "panning parameters" [92–94], "stereo panning spectrum" [31], "pangram" [95], "azimugram" [96, 97], "amplitude panning factors" [98], "pan map" [64], "pan time-frequency filter (TFF)" [84], "mixing parameters" [99, 100], and "panning coefficients"[3] [101–105].

While all the aforementioned parameters have the intention of estimating the panning that was applied to a mono stem, the exact mathematical methods vary tremendously; interested readers are directed to the original studies, although as an example, a block diagram for SPS [31] is shown in Fig. 3. Note that unlike the frequency-domain spatial cues shown in Fig. 2, the stereo spectrogram is not converted to magnitude and phase components, but instead is processed by the complex functions "channel similarity" and "partial similarity".

A general theme is that estimations of amplitude panning can be determined through ILD-related methods and measurements of "ambience" can be obtained using techniques based on correlation between the channels [29]. An interesting, but rarely-used technique is [102] in which the stereo subband signal has principal component analysis (PCA) applied to find a dominant signal and a panning coefficient is subsequently estimated using a Lagrange multiplier. Woodruff et al. and Woodruff

[3] Note that [39, 101] also use the term "panning coefficients" to describe a higher-level panning descriptor derived from a PH (see Sect. 3.2). This use of the term does not describe the same concept as the ones listed above.

and Pardo [99, 100] demonstrate the estimation of both amplitude and time panning, utilizing TF-domain ILD and IPD directly.

In all cases there is an implicit assumption that each frequency subband within the mix contains only a single direct sound source. For some use cases (e.g., source separation) this results in an underspecified problem, but for spatial *description* this is not necessarily a limitation. In fact, one can often expect recorded music to be spatially dense, and any description created from panning parameters should reflect that.

3.3.3 Time-frequency decomposition

In all the cases mentioned in Sect. 3.3.2, the panning parameters are calculated on individual TF bins. Naturally, there are several ways to perform the TF decomposition, each with its own nuances. The majority of algorithms described use the STFT as a starting point due its computational efficiency, although it is noted in several studies that the Fourier transform's lack of perceptually-relevant frequency bands is a problem and should be corrected for. Typically, fast Fourier transform (FFT) frequency bands are combined into groups according to a psychoacoustic model, such as equivalent rectangular bandwidth (ERB) or Bark critical bands [106, 107]. Tzanetakis et al. [31] takes an even more drastic approach, forming just 3 subbands (lows, mids, and highs) according to the typical practice of mixing engineers.

Li and Driessen [102] takes a similar approach but instead of using STFT, the frequency decomposition is achieved via a pseudo quadrature mirror filter (PQMF) filterbank, as defined in the MPEG audio standard [82]. Tardieu et al., De Sena et al., and Zhu et al. [42, 80, 108] choose instead to calculate stereo "cochleagrams" in the time domain; a bank of gammatone filters is applied to the original signal and the resulting subbands segmented in time. This has the benefit of being a much closer approximation to human hearing, but comes at the expense of higher computational complexity. Subsequently, it is common to apply an auditory model, i.e., a computation that models the cochlea and its neural connection to the brain. Several versions can be found through the literature, although a classical version is presented in [109], and [26, 75] include comparisons of several models.

3.3.4 Frequency-azimuth analysis

After converting audio into bin-level spatial cues, it is common to visualize them in a 2D spectrogram-like plot. A theme across most studies is that it is generally useful to view the change in spatial parameters change over time. We therefore see most often that time is displayed on the X-axis of the plot, with frequency and azimuth being assigned to either the Y-axis or colour/brightness. Further analysis of these plots can generate musical insights, as in [96] which uses changes over time to achieve song-structure segmentation of recorded music, and [62] which uses the visualizations directly to compare the rendering quality of different spatial audio formats and lossy compression algorithms. Focused purely on visualization, [40] adds the concept of bin-level "clarity", effectively determining the trustworthiness of a panning parameter estimation. As with the previously discussed spatial cues, the naming of these visualizations varies freely (see Sect. 3.3.2), and interested readers are directed to the original papers for examples.

3.3.5 High-level descriptors and perceptual scalars

Subsequent to bin-level analysis, there are several algorithms that condense the derived parameters down into a compact representation for the purpose of describing a certain spatial aspect of the original signal. In this context, "high-level descriptors" are those mentioned in Sect. 3.2, e.g., PH, ASF, PSF, and SPF.

"Perceptual scalars" are the spatial qualities often evaluated in perceptual testing, which can be predicted both from low-level spatial cues or from the high-level descriptors. Two common ones are apparent source width (ASW) and LEV. Again, there are many synonyms or near-synonyms for these terms: "stereo width", "source width", "auditory source width", "ensemble width", and "stereo image width" for ASW "envelopment", and "immersion", and for LEV.

ASW is usually calculated as a summary of either panning or DOA calculations. For example, [34] defines ASW as the full width half maximum (FWHM) of a Gaussian curve fitted to a PH, [26] uses the interquartile range (IQR) of an IID histogram, and [27] uses the standard deviation of a combination of IID and ITD histograms. Mieth and Zölzer [104] on the other hand uses IACC and lateral energy fraction (LEF) as proxies for ASW and LEV respectively. Continuing from the diagrams given in Sect. 3.3.2, Fig. 4 shows two exemplary block diagrams of algorithms to calculate ASW. Note that each algorithm contains an intermediate PH which can be used as a high-level descriptor directly. Importantly, the algorithms are completely disparate and the resulting ASWs/PHs cannot be directly compared, despite having the same name.

It has also been shown that perceptual scalars such as "immersion" can be predicted via ML from combinations of low-level features, including those which do not contain any spatial information [37]. Figure 5 shows a block diagram for this. Similarly, [110] develops a computer model of "envelopment" based on IACC, PCA,

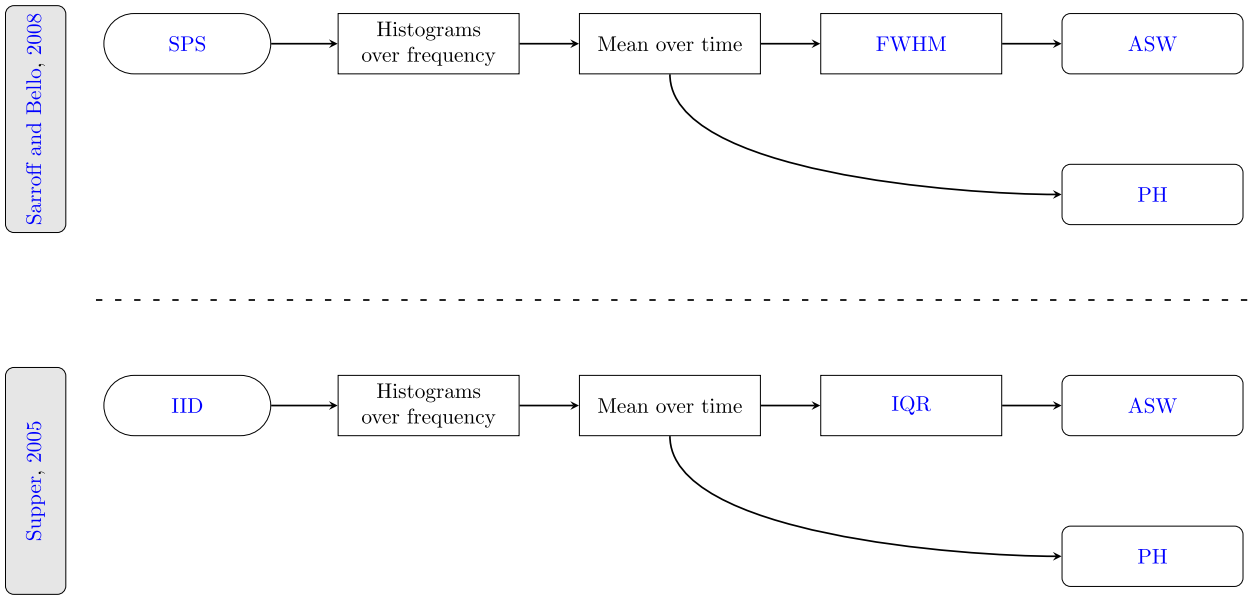


Fig. 4 Block diagrams of two algorithms to estimate ASW. Disparate algorithms are separated by a dashed line

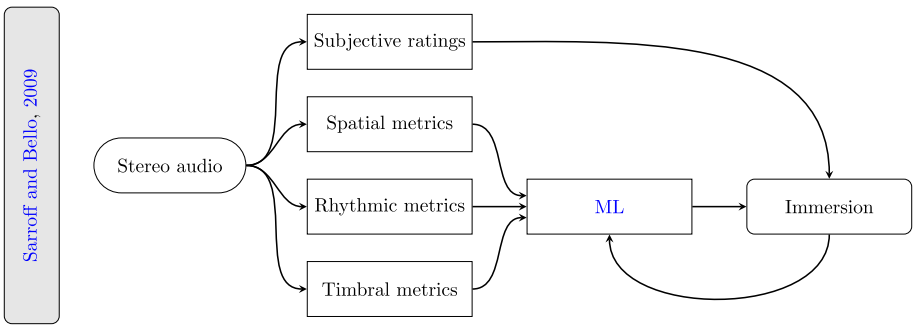


Fig. 5 Block diagram of prediction of immersion via ML on objective metrics [36]

and channel energy distribution (albeit for multichannel audio rather than stereo).

The lack of mathematical standardization in these studies emphasizes the point from Sect. 3.3.1—future research should preferably state formulas explicitly in the text, falling back to a specific definition in a cited publication.

3.3.6 Machine-learned spatial descriptors

Departing from the traditional DSP spatial descriptors, a new family of descriptors generated directly though ML and DL have emerged. As mentioned in Sect. 3.2, [13] encodes the spatial information present in multichannel audio to a vector in latent space by first downmixing the input audio to stereo and then forcing the neural network to upmix back to the original. Manocha et al. [111] takes a very similar approach, but acting on binaural input and its accompanying monophonic mixdown. In both cases, content invariance of the network is one of the optimization goals, and is essential when using the latent vectors as SIPs. The former achieves it by creating two random 5-channel mixes and using the stereo downmix of one to resynthesize the other, while the latter utilizes an additional training task which optimizes the overall network's spatial embeddings to match that of a pre-trained neural binaural decoder which takes only directional coordinates as input. Simplified block diagrams for both methods are shown in Fig. 6; the reader is directed to the original papers for details. A slightly different method is taken by [112], in which stereo audio captured with accompanying video is modified by either swapping the left and right channel, or using audio from a different video clip. The network then learns to spot said modifications, hence learning the spatial relationship between sound events and visual events.

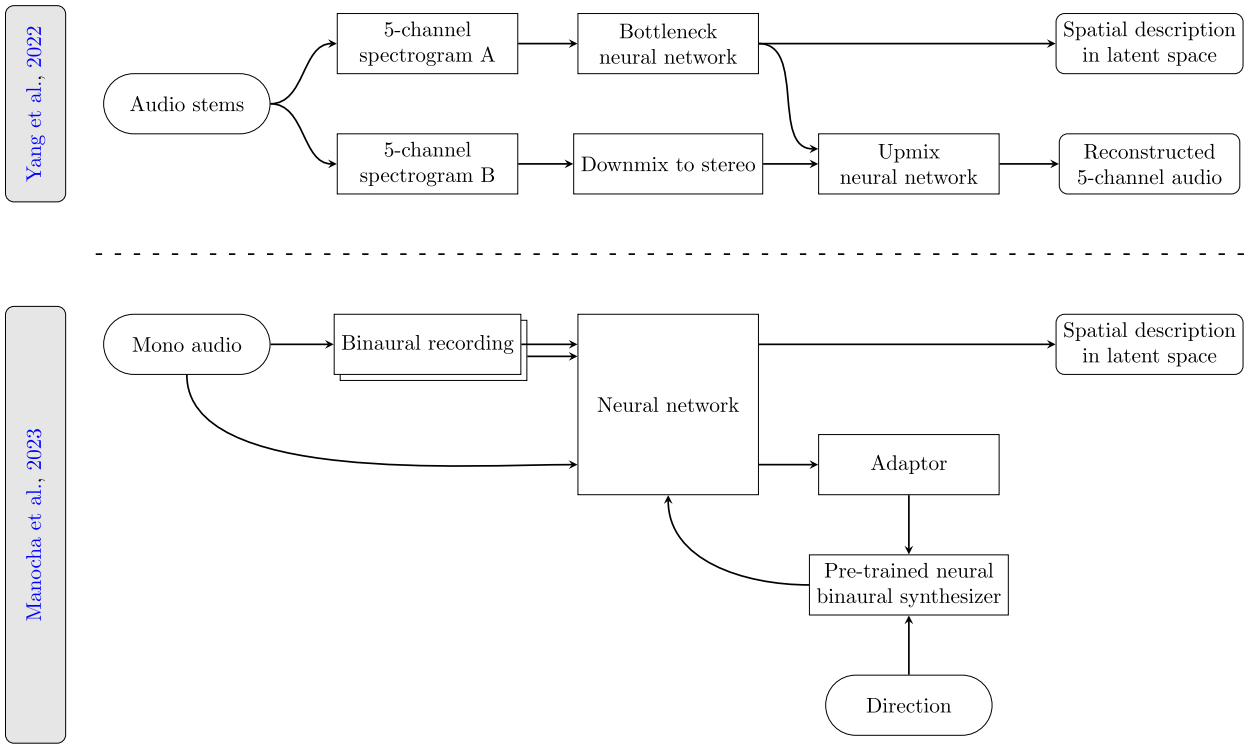


Fig. 6 Block diagrams of spatial description via DL [13, 111]

Where latent features are used a spatial descriptor, it is pertinent to remember that the latent space is a product of the training data and loss function, as opposed to the physical or psychoacoustic principals that DSP-based descriptors are typically based on. While the inherent dimensionality reduction of latent spaces is useful for comparing spatial features between recordings, individual dimensions of the space do not usually have a physical analogue. Analysis and interpretation of latent spaces is an ongoing research topic that has seen promising results in other research fields [113–115], and creating links between latent and physical space is viewed as critical to the applicability of the learned model [116].

Huang et al. and Zieliński et al. [117, 118] both attempt to learn spatial features of multichannel spectrograms using convolutional neural networks (CNNs). The former exploits an over-fitted network on short audio frames to generate inter-channel transfer functions (ITFs) in an effort to generate new coding/compression possibilities. The latter (which compares traditional DSP-based spatial features with those learned by a CNN for a binaural front-back classification task) offers a cautionary note—it found that while a CNN could produce features that resulted in higher accuracy for one specific subtask, the traditional features were more generalizable. Somewhat similarly, [119] creates estimations of fine-grained directional audio by creating neural embeddings of multichannel spectrograms generated from distributed microphone arrays.

On the side of generative ML/DL, [73] shows that a network can be trained to generate plausible Parametric Stereo (PS) (i.e., IID, ITD, and IC) values from a monophonic input (with upmixing as the intended use). More recently, Large Language Models (LLMs) have been used to generate textual descriptions of the spatial qualities of audio recordings; a good summary of the various different implementations is given in [120].

3.3.7 Standardized spatial cues

This review surfaced two spatial measurement systems that have attempted to become industry standards. The Quality Evaluation of Spatial Transmission and Reproduction using an Artificial Listener (QESTRAL) test methodology, detailed across 4 parts [121–124] is designed to measure spatial distortions present in audio processing, transmission, and reproduction. Additionally, Objective Measurements of Perceived Audio Quality-Multi Channel (PEAQ-MC) [48] is a multichannel extension to the PEAQ international standard [47], in which subband ILD, ITD, and IACC differences are calculated between a reference and distorted signal. In both cases the signals under test are expected to be in binaural

format. Other formats should be rendered to binaural by, e.g., convolution with a head-related impulse response (HRIR). It is worth noting that while the development of the measures is well documented and rooted in good theory, the review revealed only a handful of uses for each standard—[125, 126] for QESTRAL and [46] for PEAQ-MC—although this could be because simple use of a standard is typically not written about in academic literature.

There also exist several audio coding standards such as BCC, PS, MPEG-surround, spatial audio object coding (SAOC), and most recently immersive voice and audio services (IVAS) that include the calculation of spatial cues which can be directly integrated into future research (assuming one has access to encoder software). Excellent overviews of these standards and their underlying techniques are given in [127, 128].

Another avenue of standardization is the use of shared open-source software to extract spatial cues. The most established is the Auditory Modelling Toolbox (AMT) [129]—a set of MATLAB routines containing a vast number of algorithms for all aspects of human hearing, include spatial ones. Essentia [130] is a C++ framework for MIR, again containing several spatial algorithms. Released more recently, Binaspect [62] is a Python library designed specifically for extraction and visualization of spatial queues from binaural audio. The Generalized Cross Correlation with Phase Transform (GCC-PHAT) algorithm (which is a measure of DOA usually used in speech-processing, but also with examples for musical audio [52, 131, 132]) has implementations in several major programming languages [133–136]. For research involving transfer functions or impulse responses, the spatially oriented format for acoustics (SOFA) file format has emerged as the de facto standard for distributing spatially oriented acoustical data [137].

3.4 Framework for SIPs

Thus, we can construct a general model of a SIP algorithm according to Fig. 7. Recognizing that SIPs are inherently context-specific, we can consider any method that effectively summarizes bin-level spatial cues as a valid outcome. Section 3.3 details all techniques that were surfaced in this review, and for mathematical definitions the reader is referred to the original studies cited throughout. As seen in many of the reviewed studies, this framework could be applied equally to audio signals of higher order than stereo.

The diagram shows two separate paths for DSP and ML -based features respectively. It is noteworthy that these paths are essentially completely disparate, only converging at the final output stages, where there are examples of both methods being used to generate high-level spatial descriptors. In general, only one path needs to be followed to create a SIP. The lists of techniques given in each block are exemplary (not exhaustive), with their items somewhat interchangeable depending on the intended use of the SIP.

Although not shown on the diagram, it is important to choose spatial features that are relevant for the type of input audio and the intended output features. For example, if the input is standard stereophonic music and the output is intended to convey amplitude panning, then all phase-based spatial cues can be eliminated. However, if the output is intended to show LEV (even given the same stereophonic input), then phase cues cannot be eliminated. If the input audio is binaural, both phase and amplitude differences are relevant, but the frequency bands in which they are relevant differ depending on the intended output. For example, DOA only requires phase below approximately 1500 Hz [81], but other tasks should follow a binaural hearing model with more complex frequency weighting [46, 138].

3.5 Validation techniques

The papers surfaced in this review revealed several high-level categories of techniques to validate the success of the research. They tend to be closely linked to the downstream task, although some are universal. The most prevalent were perceptual listening tests, used in approximately half the studies. Some standardized perceptual tests are ITU-R-BS.1534 [139], which defines the MUlti Stimulus Hidden Reference and Anchor (MUSHRA) test used for measuring "intermediate audio quality", and ITU-R-BS.1116 [140] which is used for measuring "small impairments" in audio quality. In general though, it is more common to set up a custom subjective test asking the participants directly about the audio feature of interest. Several studies [53, 141–143] described their subjective tests as "MUSHRA-like", showing that the standardized frameworks are useful as a basis even if they do not perform the exact task required.

The second popular validation technique is objective metrics. The studies reviewed used a large variety of different metrics, although there is some standardization within specific scientific communities. For example, the blind source separation evaluation (BSS_EVAL) software library is frequently used to measure the similarity of audio waveforms in source separation tasks [144]. The same library can also be used for any task where waveform similarity is pertinent, e.g., [145] which is concerned with audio coding and compression. Another option for comparing the similarity of two waveforms is the "spectrogram difference" [146], which can be considered more in line with human perception but naturally contains a trade-off in time and frequency resolution.

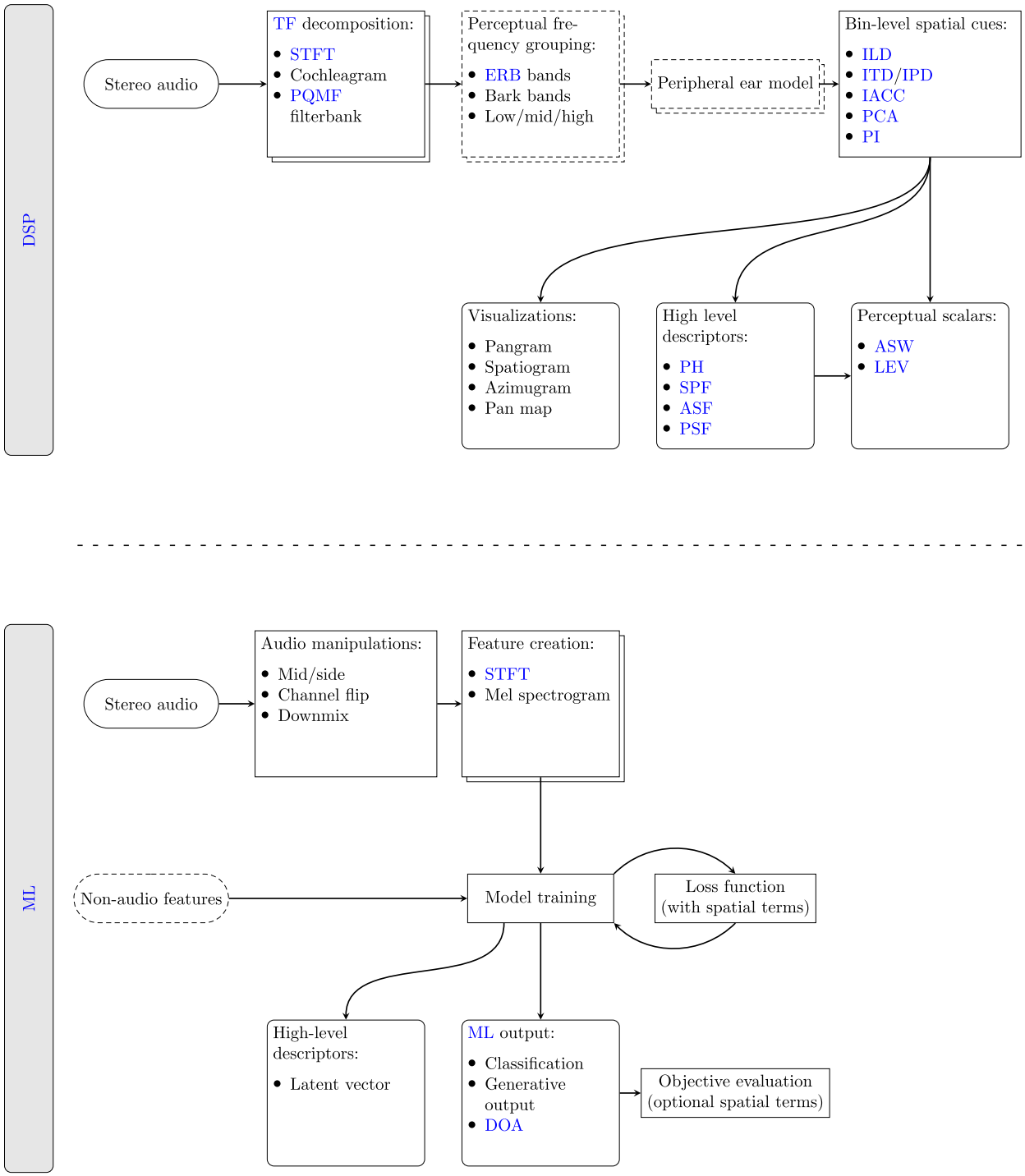


Fig. 7 Flow diagram of a generic SIP algorithm. Doubled blocks denote steps where the same processing is applied to both stereo channels. Dashed blocks are optional. The diagram is split between DSP and ML techniques, only one of which need be applied

Of note here is that the aforementioned techniques are tailored towards monophonic audio (although they can trivially be extended to multiple channels). More specific comparison metrics such as perceptual spectral difference (PSS) [147] or binaural-auditory-model-based quality (BAM-Q) [148] can be used to describe the difference between two stereo (specifically, binaural) signals. For perceptual differences in higher audio models (e.g.,

multichannel or ambisonics) it is typical to record or render the audio to a binaural signal and evaluate that [60, 149, 150].

If the research in question is related to creating or testing an objective metric, it can be useful to reverse the problem; One can perform simulations of real audio data by designing a set of signals with predictable differences, and analyze the results of the metric. Watcharasupat and Lerch [61] is one such example in which a novel metric is defined, and then its real results are tested against theoretical ones. In addition, diagrams are of the simulations are shown for visual inspection. This is another common validation technique in studies where a known, correct, low-dimensional value can be compared against a higher-dimensional metric, e.g., [38] which compares the panning position of an instrument to the peak in a PH, or [151] which compares a known azimuth and elevation with various activity maps generated from ambisonic audio. Note that comparing the similarity of PHs is error-prone as the standard distance metrics (e.g., Euclidean distance) do not necessarily correlate with perceptual difference [152]. Gomez et al. [39] attempts to side-step this issue by converting PHs into a different domain, although the efficacy of this approach remains largely unexplored.

In the ML/DL sphere it is normal to calculate the success of a model by its percentage of correct predictions, both for classical ML [31, 42, 49], and more modern DL approaches [50, 118, 153]. However, this obviously only works with labelled data. In the case that labels do not exist, one option is to create a self-supervised "pretext task" in which labelled data can be generated from unlabelled [112, 154–156]. In cases where there are no known objective metrics for the task at hand (e.g., the plausibility of novel stereo renderings [73]) inspiration can be taken from non-audio-domain tasks, utilizing, e.g., the minimum error of a large sample or the Fréchet distance between real and generated parameter spaces [157].

An under-studied but nonetheless important metric is the computational complexity of the algorithm described. Typical metrics to measure this are the time taken to process a given duration of audio on specific hardware [158], number of floating point operations (FLOPs) [159], or (for ML/DL) number of network parameters [159].

3.6 Audio datasets

3.6.1 Existing datasets

Table 5 shows a comprehensive list of audio datasets that were referenced by papers in this review. The only exception is "mobygratis" [168], which was released after the review was performed but nonetheless presents similar opportunities for research. Datasets that were not available publicly such as those used in Moving Picture Experts Group (MPEG) standardization have been omitted, and the availability of each dataset is noted. The "Size" column represents the number of musical tracks or clips present in the dataset, but does not take their lengths into account. "Duration" is given in days, hours, or minutes as appropriate, and is calculated as the sum of all track or clip durations. Non-musical recordings are not counted. Since it is assumed that larger and more varied datasets are preferable, the table is ordered by "Size". The columns "M", "S", and "B" indicated whether the datasets contain multitrack[4], stereo, or binaural recordings respectively. IEEE Signal Processing Society Online Resources, Li et al. and Bogdanov et al. [161, 182, 192] also contain lists of audio datasets (with some overlap to those listed here) that might be useful for future research.

3.6.2 Synthesized datasets

It is often the case than a dataset matching the exact task at hand does not exist, in which case one is synthesized from other datasets. In fact, some datasets listed in Table 5 are amalgamations of others—see the "Notes" column for details.

When creating a custom dataset, the most prevalent technique surfaced in this review was to convolve monophonic audio with some form of Impulse Response (IR). Said IR can be binaural (e.g., [76]), ambisonic (e.g., [193]), or a microphone array (e.g., [194]). It is also common that these the IRs add reverberation to a recording, as in e.g., [78, 195–197]. For research focused on binaural audio, [118] contains a comprehensive list of publicly available HRIR datasets.

Synthetic datasets of stereo music are slightly different in that they must be manually mixed from monophonic stems, which can be amplitude-panned in any way that the experiment requires. For example, in [44] the stems are placed according to the signal model described in the study, and [13, 198] pan the stems randomly.

Taking a slightly different route, [164] introduces a dataset of song stems synthesized entirely from Musical Instrument Digital Interface (MIDI) files using professional-grade virtual instruments (VIs), motivated by a lack of commercial music datasets offering stems.

3.7 Uses of stereo image profiles

3.7.1 Current uses

Broadly speaking, there are two categories of tasks that spatial information in audio has historically been used for. Firstly there are those focused on a single recording from which the information was extracted. Compression, source separation and upmixing are the most

[4] A Multitrack recording indicates that individual stems of a recording session are available, but not necessarily a stereo mix.

Table 5 Audio datasets, ordered by size (number of tracks or clips)

Name	Audio format	Size	Duration	M	S	B	Availability	Notes
FMA [160]	mp3 various bitrates	106574	343d		✓		Download	Real music. Huge variety. Contains common audio features and tags.
MTGJ [161]	mp3 320kbps	55609	157d		✓		Download	Real music. Wide variety of genres, instruments, and moods.
ccMixter[162]	Various mp3/wav	≈ 48000[§]	?[§]	✓	✓		Download[†]	Real music in many genres. Some stereo, some stems. Vast, but some download links are broken.
ASASCBRM [163]	wav 48kHz 24bit	5928	11h		✓		Download	Synthesized binaural from other multitrack datasets. 30 s clips.
Slakh [164]	wav 44.1kHz	2100	145h	✓	✓		Download	Mono stems. Synthesized from midi.
FAIR-Play [165]	wav 48kHz 16bit	1871	5h		✓		Download	Binaural. 10 s clips. Only 1 recording room.
MAESTRO [166]	wav 44.1/48kHz 16bit	1276	199h		✓		Download	Only classical piano.
MSSS [167]	wav 44.1kHz 16/24bit	625	40h	✓	✓		Download[⋆]	Stems uncompressed. Example stereo mixes in mp3 only.
mobygratis [168]	wav 44.1kHz 24bit	499	35h	✓	✓		Download[⋆]	Stereo stems and mixdowns. Multiple genres but only 1 artist.
RWC [169–172]	CD, 44.1kHz 16bit	465	22h		✓		CD[‡]	Multiple datasets each of a different genre.
MedleyDB 2.0 [173]	wav 44.1kHz 16bit	330	7h	✓			Download[‡]	Real music. Mix of genres. Some instrumental, some vocal. More tracks have been added since publication.
MUSDB18 [174]	AAC 256kbps	150	10h	✓			Download	Only stems. Derived from other datasets such as [175, 176].
MUSDB18-HQ [177]	wav 44.1kHz	150	10h	✓			Download	Only stems. Derived from other datasets such as [175, 176].
MedleyDB [176]	wav 44.1kHz 16bit	132	7h	✓			Download[‡]	Stereo mix + dry and processed stems.
DSD100 [175]	wav 44.1khz	100	7h	✓	✓		Download	Stems and downmixes, derived from [167].
BTPAB [178]	wav 48kHz 16bit	49	49m		✓		Download	60 s clips. Only 1 recording session.
FiloSax [179, 180]	wav 44.1kHz 24bit	48	5h	✓			Download[‡]	Only jazz.
DAOC [181]	CD, 44.1kHz 16bit	44	25m		✓		CD	Anechoic orchestra.
URMP [182]	wav 48kHz 24bit	44	1h	✓			Download	Anechoic classical music. Only stems.
BOMA [183, 184]	CD, 44.1kHz 16bit	31	1h		✓	✓	CD	Anechoic and semi-reverberant orchestra.
TSS [185]	wav 44.1kHz 16bit	23	1.5h	✓			Download[⋆]	Both live and studio sessions.
QUASI [186]	wav 44.1kHz	11	≈ 30m	✓			Download[‡]	Mix of genres.
SQAM [187]	FLAC 44.1kHz 16bit	10	1h		✓		Download	Not all music, some stereo some mono.
RIOOSSR [188]	wav 48kHz 24bit	4	5m				Download	Ambisonic and channel-based versions.
ARSSO [189]	mp3 192kbps 48kHz	4	11m	✓			mp3: Download, wav: Contact author	Anechoic classical music. Only stems.
UIowaMIS [190]	wav 44.1kHz 16bit or 96kHz 24bit	?[‖]	23h	✓			Download[†]	Isolated notes, not musical compositions. ≈ 85% stereo.
Bach10 [191]	wav 44.1kHz	?[§]	?[§]				Contact author	Only Bach chorales.

M Multitrack, *S* Stereo, *B* Binaural

[⋆]Individual tracks. [†]Individual files. [‡]Must request access. [§]Exact size unclear. [‖]Instrument recordings

prevalent examples which arose in this review. In recent years, these have become somewhat intertwined with the advent of new commercial spatial audio formats such as Dolby Atmos and MPEG-H 3D Audio; older content which was released in stereophonic or multichannel format can benefit from combinations of source separation, upmix, and coding/compression techniques that increase the perceptual quality when rendered on modern equipment. The spatial attributes extracted in these cases tend to be the low-level spatial cues detailed in Sect. 3.3.1 since the tasks naturally require that the original audio is re-synthesized in some way.

Quality prediction (and in particular, *spatial* quality prediction) is another area of considerable interest, the driving force being that traditional perceptual listening tests are slow and costly [75, 199]. This differs slightly from the previously mentioned tasks in that it implies a comparison between a reference and degraded signal, and uses summaries of low-level spatial cues combined with non-spatial measurements. The type of spatial attributes used in these cases are more in line with the high-level descriptors detailed in Sect. 3.3.5. In particular, [45, 60, 61, 200] use distortion/difference of ILD/ITD/ITD in their perceptual models, and [26, 27] use "sound field maps" or panning histograms to describe the spatial information present in the analyzed audio. Combining the two approaches, [201] defines the directional loudness distortion (DSD) metric to show the difference between a reference and degraded "directional loudness map".

The studies surfaced in this review reveal that there was a spike of interest in binaural quality prediction and analysis since 2023. This appears to be driven by a renewed interest in spatial audio[5] [75, 149, 150, 202, 203], room acoustics [196, 204] and VR rendering [72, 205, 206], along with the increased ease of gathering or creating binaural audio datasets.

Secondly, there are those tasks which focus on the relationship of a single recording to a larger corpus. Classification is a typical use case, with the list of predicted labels including genre/recording style [14, 17, 31], artist/composer/DJ [17, 207], and decade [42]. Important to note in all of these cases is that the ground truth is known a-priori. However, with the vast size of modern audio datasets, acquiring reliable annotations becomes increasingly difficult. For example, [32, 55] both specifically state that panning parameters, digital audio workstation (DAW) sessions, and musical scores are typically not available for most recorded music, and therefore any such information must be derived after-the-fact from a mixdown.

An interesting middle-ground between the two categories can be found in the aforementioned [96], which uses temporal changes in PHs to segment a song into musical structures (verse, chorus, etc.). Likewise, [43, 44] attempt both a generic classification of stereo field, and a further application to segment longer audio recordings using that classifier.

In addition, a frequently occurring pattern among studies in this review is that spatial cues are used as objective measurements for other tasks. In particular, [178, 208–210] demonstrate that the incorporation of spatial terms into DL loss functions, forcing the networks to learn or generate spatial features.

3.7.2 Potential uses

Compelling areas for further research can be inferred from what is missing from the survey results. In total, 48 studies were deemed to be making use of ML or DL, all of which were performing some form of *supervised* learning. Just a single study [14] mentioned performing *unsupervised* learning on spatial cues—a self-organising map was used to discriminate between two clusters of tracks when simple linear regression failed. Wang et al. [211] presents a similar study, although the audio was non-musical. Mitic and Rossholm [152] gives an overview of the suitability of various panning features with regard to unsupervised learning, and [62] describes some spatial cues that can be used as input to ML routines.

Likewise, relatively few studies perform any spatial analysis of extremely large and varied musical datasets. For example, [146] used music clips from FMA [160] as background noise in a task focused on the binaural rendering of speech, and while the standard spatial cues were extracted, the music was not of primary interest. Tzanetakis et al. and Tardieu et al. [17, 31, 42] use mid-size (thousands of tracks) datasets to demonstrate various classification tasks, although they unfortunately predate the existence of the aforementioned extremely large datasets. Nonetheless, these older studies present an interest in the dataset itself as opposed to low-level features of the audio. A much more recent study [155] expresses a similar interest, measuring production tends in a huge (albeit private) dataset containing hundreds of thousands of tracks. Analysis of the spatial features in vast musical corpora therefore presents an opportunity for further research. In particular, there are no studies on spatial degradation caused by lossy compression of said datasets, even though the largest [160, 161] are both encoded using the MPEG layer 3 (MP3) codec, and [62, 210] clearly show spatial degradation of their test signals

[5] As mentioned in Sect. 3.5, the typical way to measure perceptual effects of higher-order audio models is to render them to a binaural signal.

when distorted with lossy compression and source separation respectively.

Ziemer et al. and Wilson and Fazenda [212, 213] present a further basis for investigation, concluding that two objective measures of spaciousness (inter-channel cross correlation (ICCC) and SPF) did not match the results of a perceptual music similarity test. This issue warrants further investigation, since research into the SIPs of tracks in a large music corpus will naturally involve a measure of similarity.

3.7.3 Example use of SIP framework

To demonstrate the effectiveness of the framework outlined in Sect. 3.4, we will consider a hypothetical experiment on a dataset that has not yet received any investigation into stereo image. Suppose the intention was to recreate the analysis of trends in production techniques from [155], but using a large open dataset such as FMA. Said study employed a simple stereo field classification based on ILD. Each track is classified as either "mono", "narrow", "balanced", or "wide". The exact equation to calculate the stereo width is not given in the study. In order to recreate such tags for FMA we can follow the DSP framework to create a SIP as suggested in Table 6.

4 Discussion

Overall, this review paints a picture of a research landscape that is somewhat fractured. The core psychoacoustic principals and terminology is fairly well standardized, although many researchers offer their own variations on the precise implementations and inferences of both high-level spatial features and low-level spatial cues. The net result is that searching for relevant literature by *downstream task* rather than *spatial attribute* will usually be most efficient. However, this method presents diminishing returns for tasks with few previous examples or non-standard terminology (such as finding SIPs), and the opportunity of "cross-pollination" from other fields of audio research is lost. This review intended to fix the cross-pollination problem, although it turned out that the resulting scope was so large that the researcher themselves became a bottleneck. Experiments in performing audio-related literature reviews using LLMs have shown potential, although it was concluded that they are no substitute for human researchers [214].

Open-source tooling such as those mentioned in Sect. 3.3.7 go some way to towards giving researchers a shared language around spatial aspects of stereo audio. Indeed, many of the studies in this review contained links to both source code and sound samples, but precious few were still active. The move towards shared audio datasets is a step forward though, since until recently large datasets of real music were either completely unavailable or confined to private collections. With the introduction of the datasets listed in Sect. 3.6 it might now be possible for music-specific audio research to share a common footing, just as happened in the image processing community [160].

While this review aimed to provide an exhaustive list of spatial descriptors, the fact that some studies were only added after following references shows that the original search method was not foolproof. On the other hand, while the search result might not have been exhaustive, the goal of gathering at least one example of all spatial descriptors appears to have been met. In the DL sphere this claim can be questioned since each study tends to focus on an individual task using a dataset specific to that task, thereby generating spatial descriptors that are also specific to that task. While classical DSP methods comprise the majority of the included studies, ML and DL are rising sharply, indicating that the latest computing techniques and datasets are enabling research previously out of reach via DSP.

5 Conclusion

A systematic literature review (SLR) following the PRISMA guidelines was performed to find methods of describing the stereo image profile (SIP) of stereophonic music recordings. The terminology, methods, and signal processing techniques are listed in detail, and the conglomeration of data extracted from the included studies provides a framework from which custom SIP algorithms appropriate for any given task can be designed and implemented. Datasets containing suitable audio material are also listed alongside descriptions of the contained audio.

Table 6 Hypothetical use of SIP framework to recreate stereo width analysis from [155] on the FMA dataset

Component	Method	Reasoning
TF decomposition	STFT	Common and computationally efficient.
Frequency grouping	None	[155] does not give a perceptual frequency grouping for stereo width.
Peripheral ear model	None	STFT does not require a peripheral ear model.
Spatial cue	ICLD	[155] states that the stereo width classification is "based on the ICLD".
Perceptual scalar	Stereo width	Calculated by, e.g., mean of all bin-level ICLD values, quantized to 4 values.

While the focus was on stereophonic recordings, many studies are equally relevant to other audio formats such as multichannel and ambisonics. Readers can benefit from inspiration and "cross pollination" of diverse areas of audio research, and the review as a whole enables researchers and practitioners to jump-start new studies into the spatial aspects of music.

Appendix 1 Search terms

Population: "music", "musical", "audio", "sound"

Context: "stereo", "stereophonic", "binaural", "spatial audio", "ambisonic", "ambisonics", "b-format", "multi-channel"

Outcome: "panning", "information retrieval", "spaciousness", "descriptors", "spatial image", "stereo image", "spatial cues", "disentangle", "disentangling", "disentanglement", "spatial decomposition", "spatial information", "spatial descriptors", "spatial quality", "spatial analysis", "binaural model", "goniometer", "lissajous plot", "systematic review"

Negative terms: "stereo vision", "computer vision", "image processing", "stereo camera", "stereo cameras"

Specific terms: "spatial audio coding", "spatial audio object coding", "directional audio coding", "binaural cue coding", "panning spectrogram", "stereo panning spectrum", "object-based audio", "mpeg-h audio", "mpeg-h 3D audio", "dolby atmos", "sony 360 audio", "sony 360 RA", "360 reality audio", "QESTRAL"

Appendix 2 Database queries

Listing 2: Web Of Science query

```
((((TS=("music") OR TS=("musical") OR TS=("audio") OR TS=("sound")) AND (TS=("stereo") OR
TS=("stereophonic") OR TS=("binaural") OR TS=("spatial audio") OR TS=("ambisonic") OR
TS=("ambisonics") OR TS=("b-format") OR TS=("multi-channel")) AND (TS=("panning") OR
TS=("information retrieval") OR TS=("spaciousness") OR TS=("descriptors") OR TS=("spatial image")
OR TS=("stereo image") OR TS=("spatial cues") OR TS=("disentangle") OR TS=("disentangling") OR
TS=("disentanglement") OR TS=("spatial decomposition") OR TS=("spatial information") OR
TS=("spatial descriptors") OR TS=("spatial quality") OR TS=("spatial analysis") OR TS=("binaural
model") OR TS=("goniometer") OR TS=("lissajous plot") OR TS=("systematic review"))) NOT
(TS=("stereo vision") OR TS=("computer vision") OR TS=("image processing") OR TS=("stereo camera")
OR TS=("stereo cameras"))) OR (TS=("spatial audio coding") OR TS=("spatial audio object coding") OR
TS=("directional audio coding") OR TS=("binaural cue coding") OR TS=("panning spectrogram") OR
TS=("stereo panning spectrum") OR TS=("object-based audio") OR TS=("mpeg-h audio") OR TS=("mpeg-h
3D audio") OR TS=("dolby atmos") OR TS=("sony 360 audio") OR TS=("sony 360 RA") OR TS=("360 reality
audio") OR TS=("QESTRAL")))
```

Listing 3: Semantic Scholar query

```
(("music" | "musical" | "audio" | "sound") + ("stereo" | "stereophonic" | "binaural" | "spatial
audio" | "ambisonic" | "ambisonics" | "b-format" | "multi-channel") + ("panning" | "information
retrieval" | "spaciousness" | "descriptors" | "spatial image" | "stereo image" | "spatial cues" |
"disentangle" | "disentangling" | "disentanglement" | "spatial decomposition" | "spatial
information" | "spatial descriptors" | "spatial quality" | "spatial analysis" | "binaural model" |
"goniometer" | "lissajous plot" | "systematic review")) | ("spatial audio coding" | "spatial audio
object coding" | "directional audio coding" | "binaural cue coding" | "panning spectrogram" |
"stereo panning spectrum" | "object-based audio" | "mpeg-h audio" | "mpeg-h 3D audio" | "dolby
atmos" | "sony 360 audio" | "sony 360 RA" | "360 reality audio" | "QESTRAL")
```

Appendix 3 Semantic Scholar API payload

Listing 4: SS search payload. See appendix B for the search query.

```
{
  "query": <see appendix B>,
  "limit": 20000,
  "fields": "influentialCitationCount,citationCount,title,venue,abstract,referenceCount,
citationStyles,year,openAccessPdf,authors,url,fieldsOfStudy,s2FieldsOfStudy,externalIds",
  "fieldsOfStudy": [
    "Computer Science",
    "Mathematics",
    "Engineering"
  ],
  "token": null
}
```

Appendix 4 Search result filtering algorithm

Listing 5: Python pseudocode to filter duplicate database records from search results

```
def normalize_title(title):
    return "".join(char for char in title if char.isalnum()).lower()

doi_seen = set()
title_and_year_seen = set()

for record in search_results:
    norm_title = normalize_title(record.title)
    if doi != "":
        if doi in doi_seen:
            record.reject = "duplicate_by_doi"
        else:
            doi_seen.add(doi)
    elif (this_title, this_year) in title_and_year_seen:
        record.reject = "duplicate_by_title_and_year"
    title_and_year_seen.add((norm_title, record.year))
```

Appendix 5 All downstream tasks

Table 7 List of all downstream tasks and their number of occurrences

Task	Count	Very relevant
MIR	15	yes
classification	10	yes
spatial analysis	8	yes
visualization	3	yes
musicology	2	yes
compression	35	no
source separation	30	no
rendering	23	no
upmix	22	no
quality prediction	20	no
coding	13	no

Task	Count	Very relevant
perceptual model	13	no
localization	10	no
psychoacoustics	8	no
acoustics	6	no
enhancement	5	no
recording	3	no
directional segmentation	2	no
synthesis	2	no
audio format conversion	1	no
mastering	1	no
restoration	1	no
inpainting	1	no
telepresence	1	no
audio object detection	1	no
beamforming	1	no
cross-modal ML	1	no
downmix	1	no
action recognition	1	no
audio forensics	1	no

Abbreviations

AMT	Auditory modelling Toolbox
API	Application programming interface
AR	Augmented reality
ARSSO	Anechoic recording system for symphony orchestra
ASASCBRM	Database for Automatic Spatial Audio Scene Classification in Binaural Recordings of Music
ASF	Amplitude stereo features
ASW	Apparent source width
BAM-Q	Binaural-auditory-model-based quality
BCC	Binaural cue coding
BMS	Bivariate mixture space
BOMA	Bang & Olufsen Music for Archimedes
BSS_EVAL	Blind source separation evaluation
BTPAB	ByTeDance paired ambisonic and binaural
CLD	Channel level difference
CNN	Convolutional neural network
CSV	Comma separated variable
DAOC	Denon anechoic orchestral collection
DAW	Digital audio workstation
DJ	Disc kockey
DL	Deep learning
DOA	Direction of arrival
DOI	Digital object identifier
DSD	Directional loudness distortion
DSD100	Demixing secret database
DSP	Digital signal processing
ERB	Equivalent rectangular bandwidth
FA	Frequency-azimuth
FFT	Fast Fourier transform
FLOP	Floating point operation
FMA	Free music archive
FWHM	Full width half maximum
GCC-PHAT	Generalized cross correlation with phase transform
HRIR	Head-related impulse response
IAC	Interaural coherence
IACC	Interaural cross correlation
IC	Inter-channel coherence
ICC	Inter-channel correlation
ICCC	Inter-channel cross correlation
ICLD	Inter-channel level difference
ICPD	Inter-channel phase difference
ICTD	Inter-channel time difference
IID	Interaural intensity difference
ILD	Interaural level difference
ILR	Interaural level ratio
IPD	Interaural phase difference
IQR	Interquartile range
IR	Impulse response
ITD	Interaural time difference
ITF	Inter-channel transfer function
IVAS	Immersive voice and audio services
KKS	The Knowledge Foundation
LEF	Lateral energy fraction
LEV	Listener envelopment
LLM	Large language model
MAESTRO	MIDI and Audio Edited for Synchronous TRacks and Organization
MIDI	Musical instrument digital interface
MIR	Music information retrieval
ML	Machine learning
MP3	MPEG layer 3
MPEG	Moving picture experts group
MSSS	Mixing secrets for the small studio
MTGJ	Music Technology Group Jamendo
MUSHRA	MUlti Stimulus Hidden Reference and Anchor
PCA	Principal component analysis
PCO	Population, context, outcome
PEAQ	Objective measurements of perceived audio quality
PEAQ-MC	Objective measurements of perceived audio quality-multi-channel
PH	Panning histogram
PI	Panning index
PQMF	Pseudo quadrature mirror filter
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
PS	Parametric stereo
PSF	Phase stereo features
PSS	Perceptual spectral difference
QESTRAL	Quality Evaluation of Spatial Transmission and Reproduction using an Artificial Listener
QUASI	QUaero Audio SIgnals
RIOOSSR	Recordings of Italian opera orchestra and soloists in a silent room
RWC	Real World Computing
SAOC	Spatial audio object coding
SIP	Stereo image profile
SLR	Systematic literature review
SOFA	Spatially oriented format for acoustics
SPF	Stereo panning features
SPS	Stereo panning spectrum
SQAM	EBU Sound Quality Assessment Material
SS	Semantic Scholar
STFT	Short-time Fourier transform
TF	Time-frequency
TFF	Time-frequency filter
TSS	Telefunken Studio Sessions
TSV	Tab separated variable
UIowaMIS	University of Iowa Musical Instrument Samples
URL	Universal resource locator
URMP	University of Rochester Multi-Modal Music Performance
VI	Virtual instrument
VR	Virtual reality
WOS	Web Of Science

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Authors' contributions
RM conceptualized the SLR, gathered and interpreted data, formed the syntheses, and wrote the manuscript. AR performed the post-hoc data analysis and edited the manuscript. The authors read and approved the final manuscript.

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Data availability
The dataset supporting the conclusions of this article is included as an additional file. It comprises a concatenation of search results from the three databases searched, in CSV format. The columns are:

Table 8 Explanation of fields in accompanying file of database search results

Header	Explanation
`url`	The URL of the record within the database
`title`	The title of the article
`abstract`	The abstract of the article
`year`	The year of publication
`authors`	The authors of the article, in the raw format given by the search results
`source`	The publication venue
DOI	The DOI of the article
`db`	The name of the database from which the record was extracted
`db_id`	The unique identifier within the database

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The author declares no competing interests.

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